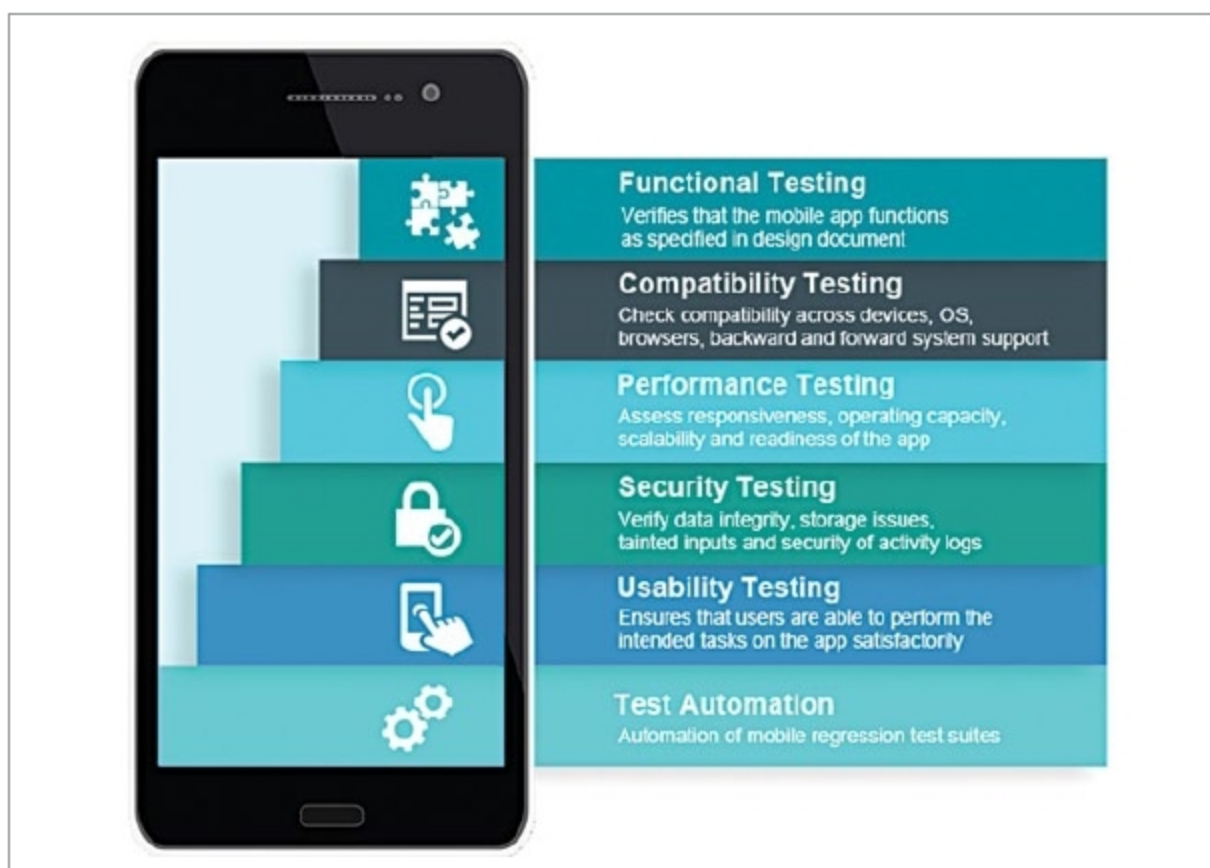




Mobile testing services (Credit: www.testingxperts.com)

and constantly update the devices to ensure data security.

Sensors use low energy but can run out of memory when loaded with many requests in the system. In an IoT environment, devices generate data with high velocity and variety. These use the IoT gateway to forward those messages to the devices, instead of directly exchanging information, which can help in balancing the load of requests received by the components. For optimised battery life and balance, these components have to be tested under various scenarios or conditions, and the right battery has to be chosen to maximise the life of the IoT devices.

Mobile testing

The challenge for mobile testers and application developers is to ensure that applications can adapt to the various types of displays on mobile devices. This means that content needs to be shown within the right boundaries so that user interaction is not hampered. Vital functions like scrolling need to be considered. To ensure mobile devices do not harm users and are compatible with devices from other manufacturers, tests have to be conducted to make sure that these satisfy all defined specifications and protocols.

A mobile device is tested thoroughly for quality before releasing to

the market. Tests that are commonly performed are mobile application, hardware, battery life, signal, network, regulatory, protocol, SAR, compatibility, pre-certification, conformance, interoperability, antenna, test result analysis and debugging support.

Safety and efficiency testing

In today's global market, products must meet numerous safety standards and industry requirements. Electrical appliances are energy-star-rated to indicate energy efficiency. Electrical safety testing is carried out to evaluate the potential risks of electrical shocks to customers while using the products. Evaluating a product for electrical safety includes tests such as high voltage, insulation resistance, leakage current and ground continuity.

An electromagnetic field (EMF) test measures the amount of electric, magnetic and electromagnetic exposure a product emits to its users, including static, extremely low frequency and RF fields. There are different regulatory limits for occupational and general public exposure. EMF testing helps meet regulatory requirements, improve product safety and reduce the risk of costly non-compliance. Third-party EMF tests and conformity assessments also help strengthen the competitive position in the market by indicating a commitment to consumer safety.

Testing by robots provides a huge opportunity for companies to prevent faulty devices from being shipped to stores from their production lines. Automation of electronics testing will help manufacturers save significantly in terms of costs and manpower. This will bring about a digital transformation in the electronics appliance testing segment.

SR-SCARA-Pro robot designed by Sastra Robotics used for testing ensures that electronic devices are well-tested before deployment towards actual usage without human intervention. This robot can integrate with renowned software quality assurance frameworks and control systems to automate testing of all software components and check against desired functionality outputs. On hardware side, it ensures that sensitivity, touch-screen intuitiveness and usability are thoroughly tested during the entire product lifecycle.

Future outlook

Integration of the IoT in consumer electronics has led to the production of more and more connected, smart and standards-compliant devices. Testing is required for these designs at every step of the product development cycle, including while designing a wireless chipset, or while integrating it into a product like a cellphone.

Test equipment that validates IoT capabilities has become an important focus for the mission-critical IoT ecosystem that must perform without failure and meet necessary compliance certification. Consumer expectations from IoT devices for reliability, performance and quality of experience are extremely high, and wireless connectivity will continue to be a critical success factor.

Each product that goes into the market needs to be tested for functionality, performance and safety. Testing in the consumer electronics industry is done at multiple stages of the design and development process. These are mostly component tests followed by integration and end-of-line production tests. Consumer electronics capable of taking autonomous decisions also need to be tested on network security, battery life and connectivity grounds. **EFY**